

Experiment No: 3

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required:

JavaScript, HTML, CSS and Code Editor.

Theory:

Control Structures and Form Validation in JavaScript

JavaScript provides control structures that allow programmers to control the flow of execution based on specific conditions and requirements. Control structures help in decision-making, repetition of tasks, and handling different program scenarios. Form validation is another important feature of JavaScript that ensures user input is correct, complete, and follows predefined rules before being processed.

1. Control Structures in JavaScript

Control structures determine how and when a block of code is executed. The main types are:

a) Conditional Statements

Conditional statements execute different blocks of code depending on whether a condition is true or false.

- **if Statement** – Executes code if a condition is true.
- **if...else Statement** – Executes one block if the condition is true and another if it is false.
- **if...else if...else Statement** – Checks multiple conditions sequentially.
- **switch Statement** – Selects one block of code from multiple alternatives.

b) Looping Statements

Loops are used to execute a block of code repeatedly until a specified condition is met.

- **for Loop**
- **while Loop**

- **do...while Loop**

These loops help reduce code repetition and improve efficiency.

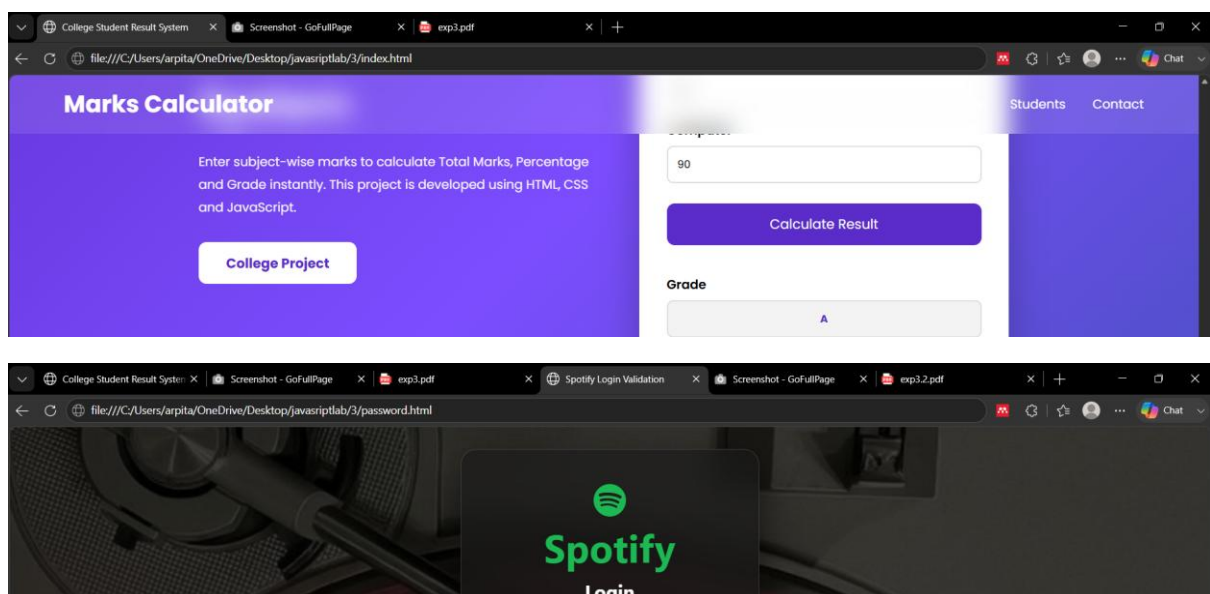
2. Form Validation

Form validation is the process of verifying that user input meets the required criteria before submission. JavaScript performs validation on the client side, reducing unnecessary server requests and improving user experience.

Common validation checks include:

- Checking for empty fields
- Validating numeric input
- Verifying marks are within a valid range
- Ensuring correct data formats
- Preventing invalid or incomplete submissions

Output



1.Case Study Program Code

Problem Statement

Design a **Student Grading System** that accepts a student's name and marks for multiple subjects. The system should validate the entered marks, calculate the total

marks, percentage, and determine the appropriate grade based on predefined grading criteria.

Implementation

The webpage consists of an HTML form where the user enters the student's name and subject-wise marks. JavaScript is used to perform **form validation**, ensuring all fields are filled and marks are within the valid range (0–100). Using **conditional statements (if...else)**, the program calculates the total, percentage, and assigns the appropriate grade. The result is then displayed dynamically on the webpage using **DOM methods** and **innerHTML**.

2. Case Study Program Code

Problem Statement

Design a **Login Form** that accepts a username and password. The system should validate the user input and ensure that the password contains at least **8 characters**, one uppercase letter, one lowercase letter, one numeric digit, and one special character before allowing successful login.

Implementation

The webpage consists of an HTML login form where the user enters a username and password. JavaScript is used to perform **client-side form validation** using **conditional statements (if...else)** and **regular expressions**. If any validation rule is not satisfied, an appropriate error message is displayed. When all conditions are met, the system displays a **"Login Successful"** message dynamically on the webpage.

Conclusion:

Control structures such as conditional statements and loops enable efficient decision-making and program flow management in JavaScript. Form validation ensures that user input is accurate and reliable. By combining these concepts, a grading system can validate marks entered by the user and determine the appropriate grade automatically, making the application both interactive and dependable.